APMA 4903 4 November 2026

The cost of a forced sale

When the book has not refilled before the next trade.

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with Allison Duffy and Jasmine Li

The reserve needs the expected bill for selling a dollar shortfall.
This section prices that sale, and checks the schedule with Q-learning.

Where this section sits

- The shortfall is $Q = \max\{L - R, 0\}$ dollars. Selling it means liquidating $X \approx Q/S_0$ shares before the call is due.
- Allison prices that liquidation with Almgren–Chriss, at risk aversion zero, and minimizes cash carry plus expected sale cost.
- Jasmine turns a sale cost per dollar, κ, into a tolerance for how often the reserve may be short.
- The bill below is the input those two formulas use. The half-life of the book is the part their refill formula leaves out.

Allison Duffy and Jasmine Li; Almgren and Chriss, 2001

Three learning problems

Supervised

Examples come in pairs: an input, and the correct output.

The error is the distance to that output.

Unsupervised

Inputs arrive with no target output.

The job is to find structure in the inputs themselves.

Reinforcement

No correct action is supplied.

The agent acts, and the environment answers with a single number.

The number scores the step. The object is the total of those numbers over the horizon, and the record of what happened depends on the actions already taken.

Sutton and Barto, 2018, §1.3

What makes the third one different

- Supervised learning can be told, after the fact, which trade would have been best. This sale cannot. The fund sees the bill of the trade it chose.
- The same state can be followed by a different bill next time, because the book and the price move.
- A cheap step can be a bad decision. Selling nothing today has no walk cost, and it leaves the whole position for the deadline.
- The data are produced by the policy. A different schedule would have left a different displacement, and the next sample would have been a different one.

Sutton and Barto, 2018, §1.3

A policy

- A policy maps the state the agent sees to the action it takes.
- It may be a formula, a table, or the result of a search. It may be random. The policy alone decides what the agent does.
- In this sale the state on a bar is the inventory still held and the displacement of the book. The action is how many lots to sell.
- The table $n_k(x, y)$ is the policy. Once it is filled, trading does not require another optimization.

Sutton and Barto, 2018, §1.3. Wording is not a quotation.

Reward, and value

- The reward is the number attached to the current step. The environment computes it from the state and the action. The agent changes it only by acting.
- This problem is written as a cost. The stage cost is the negative of a reward, and the update takes a minimum where Sutton and Barto take a maximum.
- The value of a state is the expected remaining total, from that state to the deadline.
- A step can be cheap and the state still poor, because of the states that follow. Decisions use values. The costs themselves are observed; the values are estimated from the sequence of sales.

Sutton and Barto, 2018, §1.3. Wording is not a quotation.

A model, and doing without one

- A model predicts the next state and the next cost from the current state and the action.
- Planning chooses the action by using that prediction before the step is experienced. Methods that do this are model-based.
- A model-free method never receives the prediction rule. It receives the cost that occurred and the state that came next.
- The dynamic program plans with the decay rule. Q-learning uses only the sampled cost and the sampled next displacement.

Sutton and Barto, 2018, §1.3. Wording is not a quotation.

The four objects in this sale

Object	In the forced sale
State	Day, shares still held, displacement of our own selling
Action	Lots sold on this bar
Stage cost	Cash the walk gives up, plus a price-risk penalty if risk aversion is positive
Value	Expected remaining cost through the deadline
Model	Depth q and half-life h, through $\beta = 2^{-\tau/h}$
Policy	Lots to sell in each state. The greedy table is enough to trade

The reserve uses risk aversion zero, so the stage cost on the slides below is the walk alone. The price-risk penalty is Almgren and Chriss's other objective.

The sale

Sell X shares over N days, $T = N\tau$, arrival price S_0.

$$x_0 = X, \qquad x_N = 0, \qquad n_k = x_{k-1} - x_k \geq 0.$$

The unaffected price is the Almgren–Chriss martingale, without their permanent-impact term,

$$S_k = S_{k-1} + \sigma\sqrt{\tau}\,\xi_k.$$

- The dollar shortfall converts at the arrival price, $X = Q/S_0$, as on the reserve slide.
- A fee εX and a permanent term $\gamma X^2/2$ are added to the bill. They are the same for every completed schedule, so they do not change the trades.
- The tables set both to zero. The slides leave them symbolic.

Almgren and Chriss, 2001

A book that remembers

Depth q shares per dollar. Selling n shares, starting from a displacement y, costs

$$n\Big(y + \frac{n}{2q}\Big).$$

A fraction of that displacement is still there next day,

$$\beta = 2^{-\tau/h}.$$

- h is the half-life. It is a property of the public book.
- Half a day left means $\beta = 0.5$: half of today's sale is still in the book tomorrow.
- The fund is assumed to see y before it trades.

Obizhaeva and Wang, 2013, Proposition 3

Refill is a special case

- Almgren–Chriss, at risk aversion zero, sells the same number of shares each day.
- Their temporary cost matches the walk only when the book is flat at the start of every bar. That is $\beta = 0$: the book refills before the next trade.
- Matching the two formulas gives their temporary coefficient, $\eta = \tau/(2q)$.
- The risk-neutral refill cost is then

$$E_{\min}(X) = \varepsilon X + \Big(\frac{\gamma}{2} + \frac{\eta}{T}\Big)X^2.$$

- With $\varepsilon = \gamma = 0$ and ten days, that bill is $X^2/(2qN)$. On the illustration it is 5.00 basis points.

Almgren and Chriss, 2001, eqs. (9)–(10)

Implementation shortfall

Perold's shortfall is arrival value minus cash received.

- If the trade list ignores the unaffected price, the expected shortfall equals the expected walk.
- The price-risk standard deviation is $\sigma\sqrt{\tau \sum_k x_k^2}$, divided by arrival value.
- At risk aversion zero this standard deviation does not enter the bill. It is large, because a multi-day sale still holds a large position.
- On the one-day book the best schedule accepts a bit more price risk than equal slices, in exchange for a smaller walk.

Perold, 1988; Almgren and Chriss, 2001

The bill the reserve uses

$$C(Q) = \varepsilon X + \frac{\gamma}{2} X^2 + \mathbb{E}\left[\sum_{k=1}^{N} n_k \left(y_{k-1} + \frac{n_k}{2q}\right)\right], \qquad \kappa(Q) = \frac{C(Q)}{Q}.$$

- κ is dollars of sale cost per dollar of shortfall. The tables quote it in basis points.
- Allison minimizes $cR + \mathbb{E}[C(Q(R))]$, with c the cost of holding a dollar of cash.
- Jasmine's flat-cost rule is the special case $C(q) = \kappa q$, which gives $\varepsilon^\star = \rho/\kappa$. Her ρ is Allison's c.

Almgren and Chriss, 2001; Allison Duffy and Jasmine Li

A quadratic bill has a different derivative

For $C(q) = aq + bq^2$,

$$\frac{d}{dR}\,\mathbb{E}[C(Q(R))] = -a\,\Pr(L > R) - 2b\,\mathbb{E}[Q].$$

- The flat-cost tolerance uses only the probability of a shortfall, and only when $b = 0$.
- The walk, and her permanent-impact term, are quadratic. The derivative then uses the probability and the expected shortfall.
- A single κ is still the right input to $\varepsilon^\star = \rho/\kappa$, read at a stated shortfall. The curve $C(Q)$ is the input to Allison's objective.

AI assisted, via Grok 4.7

The dynamic program

$V_k(x, y)$ is the remaining cost with inventory x and displacement y. The last trade liquidates:

$$V_{N-1}(x, y) = x\Big(y + \frac{x}{2q}\Big).$$

Earlier,

$$V_k(x, y) = \min_{0 \le n \le x} \left\{ n\Big(y + \frac{n}{2q}\Big) + \lambda\sigma^2\tau(x - n)^2 + \mathbb{E}[V_{k+1}(x - n, y')] \right\}.$$

- The expectation uses the decay rule. This is the model-based calculation.
- The reserve sets $\lambda = 0$. The variance term is then absent, and the schedule minimizes the expected walk.

The dynamic program, as a procedure

Input: depth q, decay β, risk aversion λ

1 **for** every inventory x and displacement y **do**
2 $V_{N-1}(x, y) \leftarrow x(y + x/(2q))$
3 $n_{N-1}(x, y) \leftarrow x$
4 **for** $k = N - 2$ down to 0 **do**
5 **for** every inventory x and displacement y **do**
6 **for** every legal sale $n = 0, \ldots, x$ **do**
7 $c \leftarrow n(y + n/(2q)) + \lambda\sigma^2\tau(x - n)^2$
8 $y' \leftarrow \beta(y + n/q)$, plus a refill shock if the book is noisy
9 score the sale by $c + \mathbb{E}[V_{k+1}(x - n, y')]$
10 $V_k(x, y) \leftarrow$ the smallest score, and $n_k(x, y)$ the sale that attains it

Line 8 is the model: the next displacement is computed from the decay rule. The reserve sets $\lambda = 0$, so line 7 is the walk alone. Line 9 then adds the expected cost of whatever inventory is left.

AI assisted, via Grok 4.7

Two limits the grid has to match

- $\beta = 0$ and $\lambda = 0$: equal trades, and the bill $X^2/(2qN)$. This is Almgren–Chriss, and Bertsimas and Lo for a quadratic temporary cost.
- $\beta = 0$ and $\lambda > 0$: the hyperbolic holdings. On the illustration, the reference penalty raises the refill bill from 5.00 to 8.12 basis points. The reserve does not use that penalty.
- A finite half-life and $\lambda = 0$: more shares on the first day and the last day than on the days between. That shape is Obizhaeva and Wang.
- If the book never refills, every completed schedule costs $X^2/(2q)$.

Almgren and Chriss, 2001; Bertsimas and Lo, 1998; Obizhaeva and Wang, 2013

The state the learner sees

- Inventory is an integer number of lots. Displacement is rounded to a grid of nodes, from a flat book up to a one-shot walk.
- The dynamic program and the learner solve that same finite Markov decision process.
- The action value $Q_k(x, y, n)$ is the cost of selling n lots now and acting optimally afterward.
- In this section Q_k is that action value. In the reserve, Q is the dollar shortfall. They do not appear in one equation.

AI assisted, via Grok 4.7

Q-learning, filled from the deadline

Watkins's update replaces the expectation by a sample, with step size $1/(\text{number of visits})$:

$$Q_k(x,y,n) \leftarrow Q_k(x,y,n) + \alpha\Big(c + \min_{n'} Q_{k+1}(x',y',n') - Q_k(x,y,n)\Big).$$

- The last bar has no continuation. Its value is the observed cost of selling what remains.
- Each earlier bar is sampled only after the next bar has been filled, so the target has stopped moving.
- With no refill shock, one visit to each state and action is the conditional expectation. The sweep reproduces the dynamic program.
- The learner is not given β or q. It is given the sampled cost and the sampled next displacement.

Watkins and Dayan, 1992. Finite-horizon sweep: AI assisted, via Grok 4.7.

Q-learning, as a procedure

Input: sampled costs and next displacements, not β and not q

1 set every action value to $+\infty$
2 **for** every inventory x and displacement y **do**
3 sell what remains, and store the observed cost as $Q_{N-1}(x, y, x)$
4 **for** $k = N - 2$ down to 0 **do**
5 **for** every x, y, and legal sale n **do**
6 take the sale; observe the cost c and the next displacement y'
7 $Q_k(x, y, n) \leftarrow c + \min_{n' \leq x-n} Q_{k+1}(x - n, y', n')$
8 $n_k(x, y) \leftarrow \arg\min_n Q_k(x, y, n)$

Line 6 replaces the decay rule. Several samples of one state and action are averaged, which is Watkins's update with step size $1/(\text{number of visits})$. With no refill shock, one sample is the conditional expectation, and the sale on line 8 is the dynamic program's sale.

Watkins and Dayan, 1992. Procedure: AI assisted, via Grok 4.7.

From the table to a trade

start from the full shortfall, $x \leftarrow X$, and a flat book, $y \leftarrow 0$
for $k = 0$ to $N - 1$ **do**
  read today's displacement y
  sell $n \leftarrow n_k(x, y)$ lots
  on the last day, sell whatever inventory remains
  $x \leftarrow x - n$
  take the displacement the book returns as tomorrow's y

- The same loop plays either table. Only $n_k(x, y)$ changes.
- The dynamic program filled its table from β and q. The learner filled its table from the samples.

Why the horizon is filled backward

- The same update, run forward from an unfinished table, treats an action that has never been tried as free, because the minimum prefers a zero that has not been sampled.
- That bias stays until every legal successor has been visited. On this grid it does not clear in a practical number of episodes.
- Backward order removes it. A bar is sampled only after every legal action on the next bar has a value.
- Watkins and Dayan prove almost-sure convergence for a discounted infinite horizon. The claim used here is the finite-horizon argument above. The dollars are not discounted.

Watkins and Dayan, 1992. Finite-horizon argument: AI assisted, via Grok 4.7.

Why this algorithm

- The fixed point of the update is the minimizer of the Bellman equation. That is the claim we need: the schedule is optimal for this process when the decay rule is sampled rather than supplied.
- A deep Q-network stores the action value in a neural net and trains it with a discounted target. The net is what makes the method run when the state will not fit in a table, as in Ning, Lin, and Jaimungal. Here the state is two integers. A net would replace the equality with the dynamic program by an approximation error.
- Proximal policy optimization updates a parameterized policy from sampled returns. It does not build the action value, so there is no cell to compare with the dynamic program.
- Soft actor-critic adds the entropy of a continuous action. That is a different objective, and the action here is a whole number of lots.

Mnih et al., 2015; Schulman et al., 2017; Haarnoja et al., 2018

What is gained past Almgren–Chriss

- Almgren–Chriss is the optimum under one model: impact is gone before the next trade, and the temporary coefficient η is known.
- If that model is wrong, the formula still quotes 5.00 basis points. On a one-day half-life the sale costs 10.81.
- The dynamic program repairs the model, provided the half-life and the depth are known and correct.
- Q-learning repairs the schedule when those two numbers are not handed over. The observed walk and the observed displacement replace them.
- In these runs the observations are drawn from the book the dynamic program uses, so the two can be compared. They agree to the cent. That agreement is the evidence. It is not a second bill.

Hendricks and Wilcox, for scale

- They also fill a table with Watkins's update.
- Their action multiplies a fixed Almgren–Chriss slice by a factor between 0 and 2. The book is assumed to have recovered before the next bar.
- The action here is the number of lots. The state their table does not carry is the displacement those lots leave behind.
- The extension is that state. The learner is then in a position to sell less into a book it has already pushed down.

Hendricks and Wilcox, 2014

Letters that are shared with the other sections

- Almgren–Chriss λ is risk aversion, in units of one per dollar. The reserve sets it to zero.
- Jasmine's λ is the Poisson rate at which a buyer appears for the private stake.
- Jasmine's ρ, and Allison's c, are the cost of holding cash.
- The speed of the public book is the half-life h, reported as $\beta = 2^{-\tau/h}$.
- Her crisis slows the Poisson rate. The downturn below changes the public book the shares are sold into.

Allison Duffy and Jasmine Li; Almgren and Chriss, 2001

The illustration

- $Q = 5$ million dollars, sold as $X = 100{,}000$ shares at $S_0 = 50$. One basis point is 500 dollars.
- Ten trading days. Normal depth $q = 200{,}000$ shares per dollar. Daily volatility one dollar.
- Selling the whole block at once walks the mid by 50 cents. The average fill is 25 cents worse: 50 basis points.
- Risk aversion, the fee, and permanent impact are zero. Then C is the expected walk and $\kappa = C/Q$.
- Grid: 40 lots and 41 displacement nodes. These rows are `execution/report.py`.

Cost per dollar of shortfall

	κ, basis points			
Half-life	Her formula	Equal trades	Optimal	Learner
within the day	5.00	5.00	5.00	5.00
0.5 day	5.00	7.25	7.25	7.25
1 day	5.00	11.50	10.81	10.81
2 days	5.00	20.25	17.94	17.94
5 days	5.00	32.75	27.50	27.50
10 days	5.00	39.50	34.25	34.25

Her formula assumes the book refills, so it does not depend on the half-life. The learner repeats the optimal column. At a half-day the grid does not improve on equal trades.

One day of memory

- Fraction left overnight: $\beta = 0.50$.
- Refill formula: 5.00 basis points, 2,500 dollars.
- Equal slices in the book that actually persists: 11.50 basis points, 5,750 dollars.
- Best schedule: 10.81 basis points, 5,406 dollars. Per dollar of shortfall, $\kappa = 0.001081$.
- The larger gap is persistence, 5.00 against 10.81. The smaller gap is reoptimization, 11.50 against 10.81.
- Price-risk standard deviation: 337.64 basis points for equal slices, 356.72 for the best schedule.

The sweep on that book

- Before any earlier bar is sampled, the greedy rule sells nothing until the deadline. The bill is the one-shot walk, 25,000 dollars.
- Each completed bar lowers that cost. With no refill shock, the last point equals the dynamic program.
- On the fine grid the curve ends at 5,406 dollars. The gap is zero.
- The horizontal line on the screen is that one number. It is the optimum of the persistent book, not the 2,500-dollar refill formula.
- The column headed Learner is this same check, on every half-life.

The public book in a downturn

Same shortfall, same ten days, same risk aversion. Depth halved, half-life five days, volatility two dollars.

	κ, basis points			
Downturn book	Her formula	Equal trades	Optimal	Learner
refill assumed	10.00	10.00	10.00	10.00
half-life 5 days	10.00	65.50	55.00	55.00

- At risk aversion zero, volatility does not enter C. Halving the depth doubles the refill formula, from 5 to 10.
- The half-life takes the best schedule to 55.00 basis points, 27,500 dollars, $\kappa = 0.0055$.
- The same half-life at the original depth costs 27.50. The ratio is the ratio of the depths.

The cost per dollar grows with the shortfall

	Optimal κ, basis points		
Shortfall	Her formula, 1 day	Optimal, 1 day	Optimal, downturn
1.25 million	1.25	2.70	13.75
2.5 million	2.50	5.41	27.50
5 million	5.00	10.81	55.00
10 million	10.00	21.62	110.00

Doubling the shortfall doubles κ. The walk is quadratic, so a constant κ belongs to one shortfall. Allison's expectation uses the column.

What is handed across

- To the reserve formula that assumes refill: replace $(\eta/T)X^2$ by the expected walk of the best schedule. Add $\varepsilon X + (\gamma/2)X^2$ when those parameters are chosen.
- To the flat-cost tolerance: $\kappa = C(Q)/Q$ at a stated Q and a stated book. At five million dollars, 0.001081 in the normal one-day book and 0.0055 in the downturn book.
- To the day the reserve is short: sell by the feedback rule $n_k(x, y)$, not by the equal-trade list.
- Q-learning is the check that this rule can be obtained from observed sales. The number in the reserve is C, and on these runs the learner and the dynamic program return the same C.

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The math note is `notes/execution-math.tex`. The screen is `streamlit run execution/app.py`.